

Practical No. 1.

In the context of blockchain technology, transactions are the fundamental building blocks that enable the transfer of digital assets and information across the network. Here are some commonly encountered classes in blockchain.

Payment Transactions: These transactions involve the transfer of cryptocurrencies or digital tokens from one address to another. Payment transactions typically include the sender's address, the recipient's address, and the amount being transferred. They are commonly used for peer-to-peer transactions, online purchases, and remittances.

Smart Contract Transaction: Smart Contract Transactions are executed on blockchains that support programmable functionality such as Ethereum. Smart Contracts are self-executing contracts with predefined rules encoded within them.

Token Creation Transactions:- These transactions are used to create new digital tokens or cryptocurrencies on a blockchain platform that supports tokenization, such as Ethereum ERC-20 standards.

Asset Exchange Transactions:- These transactions facilitate the exchange of digital assets, including cryptocurrencies, tokens, or other digital representations of real-world assets.

Data Transactions:- Some blockchain allows for the storage and retrieval of data on the blockchain itself. Data transactions involve storing or updating data on the blockchain, so

Governance Transactions:- In blockchain networks with the governance mechanism, such as decentralized autonomous organizations, governance transactions are used to participate in decision-making process.

Practical No. 2a.

Mining Ether refers to the process of validating and adding new transactions to the Ethereum blockchain while also securing the network.

Proof of Work :- Ethereum currently uses a consensus mechanism called Proof of Work (PoW) for mining Ether. Miners compete to solve complex mathematical puzzles to validate transactions.

Mining Equipment. To mine Ether efficiently, miners typically use specialized hardware called graphics processing units or application-specific integrated circuits.

Mining Pools :- Due to the high computational power required to mine Ether, many miners join mining pools. Mining pools are groups of miners who combine their resources to increase their chance.



Block Rewards:- Miners are rewarded with Ether for successfully mining and adding new blocks to the Ethereum blockchain. As of the knowledge cutoff in September 2021, the block reward for mining Ether's

Transaction Fees:- In addition to block rewards, miners also receive transaction fees paid by users for including their transactions in the blocks they mine.

Difficulty Adjustment: The Ethereum network automatically adjusts the difficulty level of the mining puzzle based on the overall computational power of the network.

Practical No. 2B

Bitcoin Core is the reference implementation of the Bitcoin protocol, which serves as the backbone of the Bitcoin blockchain. Developed by the Bitcoin Core project, it is an open-source software that provides a full node implementation for the Bitcoin network.

Full Node:- Bitcoin Core operates as a full node, meaning it maintains a complete copy of the entire Bitcoin blockchain. It verifies and validates all transactions and blocks, ensuring the network's security and integrity.

Wallet Functionality:- In addition to being a full node, Bitcoin Core includes a built-in wallet. Users can generate and manage their Bitcoin addresses, send and receive transactions, and store their private keys securely.

Network Communication: Bitcoin core node communicate with each other using the Bitcoin protocol. They exchange information about transactions, blocks and network updates through a peer-to-peer network. By participating in the network communication.

Consensus Mechanism: Bitcoin core follows the Proof of Work (PoW) consensus mechanism, where miners compete to solve cryptographic puzzles to validate transactions and create new blocks.

Security and Privacy: Bitcoin Core implements various security measures to protect users funds and maintain privacy. It supports the use of private keys, encrypted wallets, and secure network communication.

Practical No. 2 C

Running a blockchain node is a fundamental component of participating in a decentralized blockchain network. It involves maintaining a copy of the entire blockchain ledger and actively participating in the network consensus.

Blockchain Validation: A node is responsible for validating transactions, ensuring they adhere to the network's rules and consensus algorithm. This validation process helps maintain the security and trustworthiness of the blockchain.

Full Copy of the Blockchain: Each node maintains a complete copy of the blockchain, which contains all historical transactions blocks. This ensures that the node has access to the entire transaction history and can verify the current state of the network.

Consensus Participation: Nodes participate in the consensus mechanisms by the blockchain network. Depending on the consensus algorithm employed, nodes contribute their computational power, storage capacity or stake to reach agreement on the validity.

Network Communication: Nodes communicate with each other through peer-to-peer protocols. They exchange information about transactions, blocks and network updates.

Synchronization: Nodes synchronize their copy of the blockchain with other nodes to maintain consistency. They exchange information and verify that their copy matches the majority of the network.

Procedure No. 43

Withdrawal Pattern: The withdrawal pattern is a design pattern commonly used in smart contracts to enable users to withdraw their funds.

Restricted Access: Restricted access refers to controlling the premissing and access. It refers to controlling the if modifiers to restrict certain functions or data to authorized individuals or entities.

Contracts: Contracts are the fundamental building blocks of smart contracts in blockchain development. Contracts contain the code, data, and functions that define the behaviour and logic of a blockchain.

Inheritance: Inheritance is a programming concept that allows smart contracts to inherit properties and functionalities from other contracts.

Constructors: Constructors are special functions within smart contracts that are executed when a contract is deployed.

Abstract Contracts :- Abstract contracts serve as template or blueprints for other contracts. They contain unimplemented functions and are intended to be inherited by child contracts.

Interface :- Interface provide a way to define common function signature that contracts must adhere to. They specify the required function without providing the implementation details.

Library :- Libraries are reusable pieces of code that contain functions and data structures. They can be deployed separately from contracts and allow contracts to call their functions.

Assembly :- Assembly is a low-level language used in blockchain development to write code at a more granular level.